Quality Measurement for Transmitted Audio Data Using Distribution of Sub-band Signals

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***Abstract*—Recently, the remarkable progress of network technology has increased the requirement for transmission of high quality multimedia data. By the trend, it has been issued to investigate an efficient methodology for quality measurement of transmitted multimedia data. In this paper, we propose a new audio quality measurement technique to substitute for a typical quality measurement tool, RMSE (Root Mean Squared Error). The proposed method modifies the variance of sub-band signals to perform the estimation of audio quality at the transmitter, the receiver is able to estimate the quality distortion of transmitted audio data by calculating the distance between the variance and the reference value representing the characteristics of sub-band signals, so called EVE (Estimated Variance Error). The proposed is as no reference technique, it does not require the original data to measure the audio quality. On the Gaussian noise channel with several standard deviations, we prove that the proposed scheme has good performance, and it is a novel alternative to RMSE.**

I. Introduction

With the rapid growth of the Internet, most consumers have requested for service providers to transmit the multimedia data with high quality. It encourages the researchers to develop the compression technology such as MPEG (Motion Pictures Experts Group) and JPEG (Joint Photographic Experts Group)/JPEG-2000, the network environment to guarantee the QoS (Quality of Services), and so on. Although the well-constructed network environment is most important to satisfy the demands of consumers, it costs enormous expense since it is primarily based on the physical layer of OSI (Open Systems Interconnection) seven layers. For that reason, recently, it has become a new issue to research schemes for quality measurement of multimedia data on the application layer. It is a good methodology to solve the cost problem.

Several quality measurement techniques of multimedia data have introduced by [1-5]. R.Reibman [3] proposed the quality monitoring techniques of video. Compared with other researches, he introduced an especial method. The proposed algorithm on [3] measures the quality distortion from the video bit-stream instead of the samples. Through the simulations, he claimed that his proposed algorithm using bit error is related on the real quality of video. In [1-2], they estimate the quality distortion of audio and still image by using fragile watermarking. The proposed embeds watermark information into frequency components, which is obtained by DCT (Discrete Cosine Transform) or DWT (Discrete Wavelet Transform). The measurement tool estimates the quality distortion from correct or incorrect extracted watermark bits. Even if the fragile watermark is broken proportional to channel error ratio, the embedded watermark could be easily distorted by unintentional error such as compressions. Moreover it is difficult to decide a feature to embed watermark, which is sensitive to quality distortion.

In this paper, we propose a new audio quality measurement technique. The proposed uses the distribution of sub-band signals, which is obtained by DWT. At the transmitter, we alter the variance of sub-band signals into the constant value. The receiver, audio quality monitor, calculates the variance error from the constant value. We denominate it as EVE (Estimated Variance Error). In general, since the variance moves according to changing of signals, we could estimate the quality distortion through checking the distance of variance. And, modifying the specific sub-band signals can decrease the distortion caused by the process of transmitter since modifying the variance of whole sub-band signals is waste of resource. Besides, it allows us to select the optimal sub-band, which is most sensitive to noise. Note that a quality measurement system should be able to estimate the quantity of noise from the transmitted multimedia data. Moreover, the proposed is no reference quality measurement technique, that is, it does not require the original audio data on the receiver.

In section II, we describe our quality measurement technique using the distribution of sub-band signals in more detail. Section III shows the simulation results. To demonstrate the efficiency of the proposed method, we compare the result with RMSE (Root Mean Square Error) on Gaussian noise channel. Finally, we conclude this paper in section IV.

II. The Proposed Quality Measurement Scheme

In transmitting the multimedia data through wired/wireless network, the quality is almost dependent on channel noise. Especially, if the characteristic of channel is represented by a kind of additive noise, RMSE (Root Mean Squared Error) can be used as a good quality measurement tool. RMSE can be redefined by the difference of variance between the original data and the transmitted data. It is proved in appendix. In this paper, we propose a quality measurement technique for transmitted audio data using the variance of sub-band signals, which is obtained by DWT (Discrete Wavelet Transform). Fig. 1 shows the proposed quality measurement system. The

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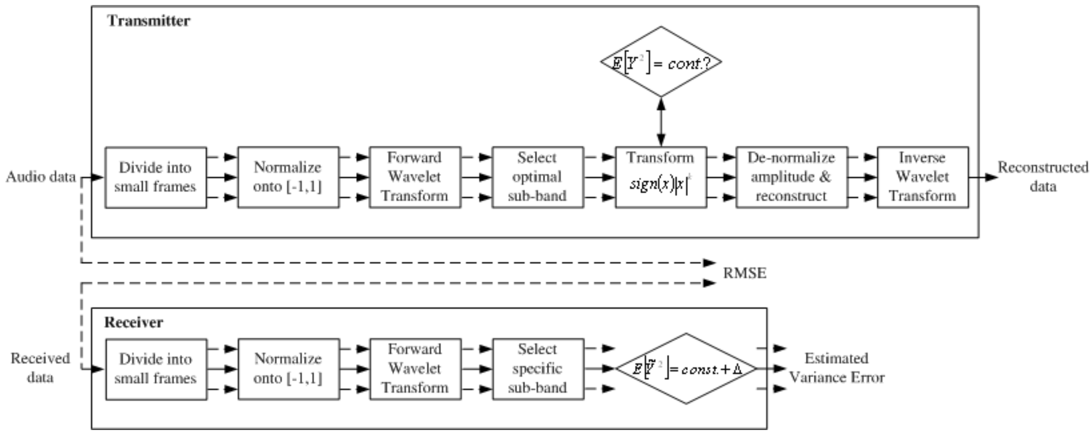

Fig. 1. The proposed quality measurement system for transmitted audio data

proposed method modifies the variance of sub-band signals into constant value, so called reference value, and estimates the quality distortion from the distance between the variance of transmitted audio data and the constant value. The reason of employing the sub-band signals for modifying the variance is that altering the whole frequency band signals has an influence on the distortion of audio quality.

Fig. 2 shows the main idea of proposed scheme. For the sake of simplicity, let's assume that the sub-band signals is a random variable, X, with uniform distribution as shown in Fig.2-(a). The distribution is modified according to reference value. For bigger reference value, the sub-band signals are modified to make their distribution concentrate on the mean value as shown in Fig.2-(b). On the other hand, for smaller reference value, the distribution becomes more distance from the mean value as shown in Fig.2-(c). The variance of random variable X with uniform distribution, σ_X^2, is as follows.

$$\sigma_X^2 = E\left[X^2\right] = \int_{-1}^{1} x^2 p_X(x)\,dx = \frac{1}{3}. \tag{1}$$

That is, if we know the characteristics of signals, the reference value can be determined as constant value. The quality measurement is quite easily performed, as the quality distortion can be estimated just by calculating the distance of variance from reference value. It means that the proposed can measure the audio quality without the original data. It is greatly correlated with RMSE. Therefore, we designate the estimated RMSE as EVE (Estimated Variance Error)

A. Transmitter; Modifying the Variance of Sub-band Signals

To measure the audio quality at the transmission side, we modify the variance of sub-band signals of original data into reference value. The original signal is decomposed into several sub-band signals by using DWT. Fig. 3 shows the procedure of modifying the variance of original audio data into reference value. Let's consider only the simple system decomposed into two channel. Firstly, the original signal $x[n]$, $(0 \le n < N$, where, N is the number of audio samples) is divided into M small frames, f_i $(0 \le i < M$, where, M is the number of frame). The signal of each frame is converted onto normalized range [-1,1]. The normalized j-th sample of i-th frame, $\hat{x}_i[j]$, is obtained by

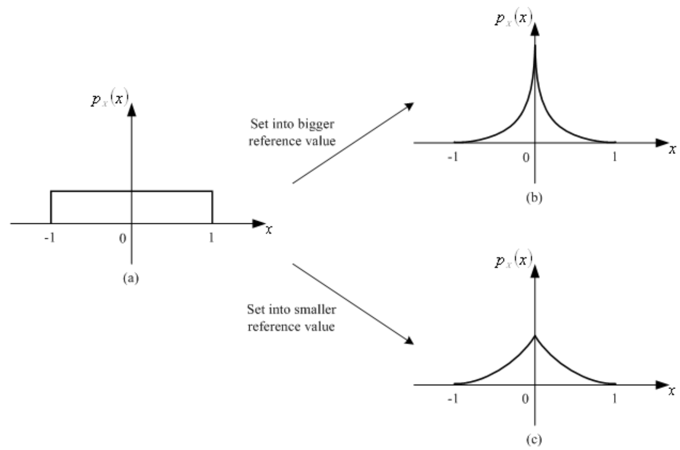

Fig. 2. Main idea of proposed scheme

$$\hat{x}_i[j] = 2 \cdot \frac{x_i[j] - \min}{\max - \min} - 1. \tag{2}$$

Where, max is the maximum value and min is the minimum value that an audio sample can have. And then, each frame is decomposed into low frequency and high frequency band signals by analysis filter bank, $H_L(z)$ and $H_H(z)$. To alter the variance of sub-band signals, one specific sub-band signals are selected. Note that, it is more reasonable that the high frequency band is selected, since its signals are more sensitive than the low frequency band signals to noise.If the sub-band signals are divided into more and more small frames, the distortion caused by modifying the variance can be decreased. However, some modified sub-band signals belong to certain frame can invade into other neighbor frames. For that reason, the sub-band signals are modified by an exponential function, $sign(x)\,|x|^k (k \in \mathbf{R}^+)$. The exponential function can cope with interference between altered sub-band signals. Fig. 4 shows that that the exponential function is able to prevent

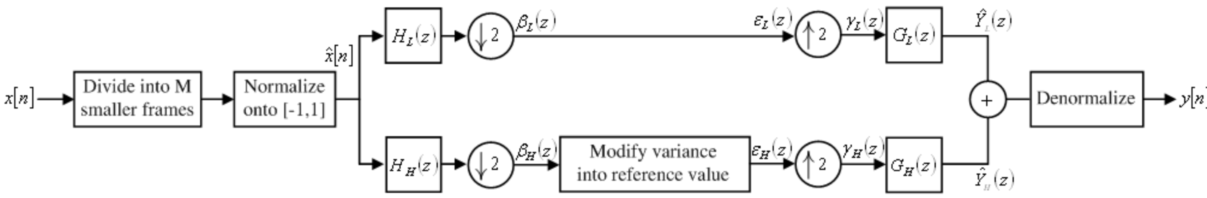

Fig. 3. The process of modifying the variance of sub-band signals

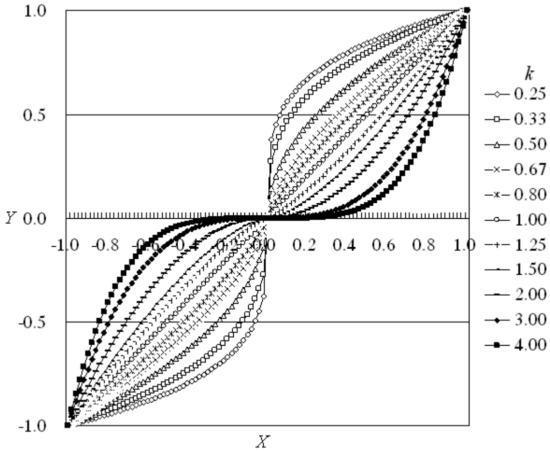

Fig. 4. The curves of modified vertex norms of bin via being transformed by exponential function in terms of k

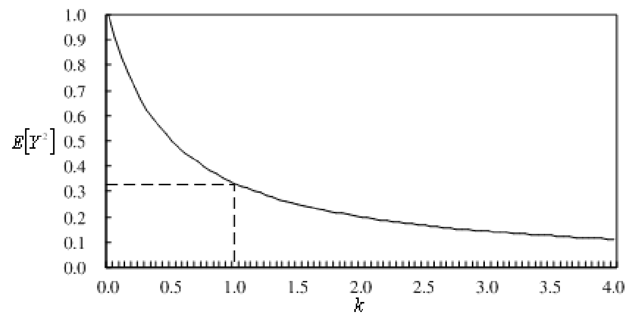

Fig. 5. The variation of variance in terms of k

the modified sub-band signals from getting over the range [-1,1]. If the modified sub-band signals are denoted by random variable Y, the variance of Y, σ_Y^2, is obtained by

$$\begin{aligned}\sigma_Y^2 &= E\left[Y\right] = E\left[\left(sign\left(\hat{X}\right)\cdot\left|\hat{X}^k\right|\right)^2\right] \\ &= E\left[\hat{X}^2\right] = \int_{-1}^{1} \hat{x}^{2k} p_{\hat{X}}\left(\hat{x}\right) d\hat{x} = \frac{1}{2k+1}. \quad (3)\end{aligned}$$

It means that we can alter the variance by controlling k. The variation of variance according to k is shown in Fig. 5. The selected sub-band signals are modified by the exponential function, and the variance become reference value, r, where, r could be determined by the characteristics of signals. For example, as shown in section II, the case of uniform distribution has the reference value of 1/3. Now, the variance of specific sub-band signals in every frame is equal to the same as reference value. Whole sub-band signals including some part of modified sub-band signals are transformed into the modified signals, $\hat{y}_i[j]$, passing through reconstruction filter bank, $G_L(z)$ and $G_H(z)$. Finally, the reconstructed audio signal, $y[n]$, is denormalized onto original range and it is transmitted to receiver. The denormalized j-th sample of i-th frame, $y_i[j]$, is obtained by

$$y_i[j] = \frac{1}{2}\cdot\left(\left(\hat{y}_i[j]+1\right)\left(\max-\min\right)\right) + min. \quad (4)$$

B. Receiver; Quality Measurement for Transmitted Audio Signal

Fig. 6 shows the proposed audio quality measurement process. It is quite simple since the quality of transmitted audio signal can be estimated just by checking the difference between the variance of the sub-band signals and the reference value, r. Similar to the process of transmitter, audio signal passing the noise channel $\tilde{y}_i[n]$, is divided into M small frames, f_i. The signal of each frame is converted onto normalized range [-1,1]. And then, each frame signal is decomposed into sub-band signals by using the same analysis filter bank as used in transmitter. In the same sub-band as transmitter of each frame, the quality of transmitted audio signal is determined as the estimated variance error, so called EVE. The EVE of i-th frame, ϵ_i, is calculated by

$$\begin{aligned}\epsilon_i &= \sqrt{\sigma_{\hat{Y}_i}^2 - r} \\ &= \sqrt{E[\hat{Y}_i] - r} \quad (5)\end{aligned}$$

Note that the proposed audio quality measurement system does not require the original data.

III. Simulation Results

The simulations are carried out on mono pop, rock, and classic music with 16-bits/sample and sampling rate of 44.1KHz, respectively. For sub-band decomposition, packet 5/3-tap bi-orthogonal perfect reconstruction filter bank [6] is applied recursively to low and high frequency band signal. Here, original audio signal is decomposed into eight sub-bands (three multi-resolution levels). The filter bank can be implemented by fast operation algorithm, called lifting [7]. The frequency ranges of each sub-band are listed in table 1. One frame consists of 4.644 sec (204,800 sam-ples) in time domain.

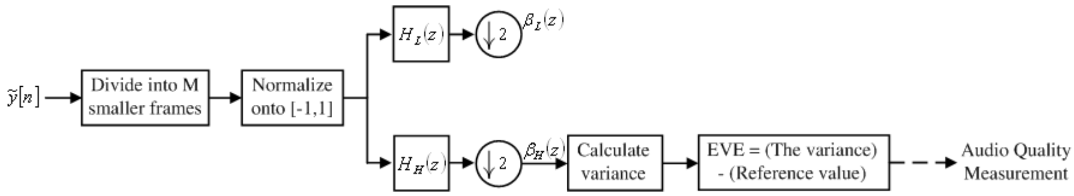

Fig. 6. The process of quality measurement for transmitted audio signal

TABLE I
FREQUENCY RANGES OF SUB-BANDS (KHZ)

1st Sub-band	2nd Sub-band	3rd Sub-band	4th Sub-band	5th Sub-band	6th Sub-band	7th Sub-band	8th Sub-band
$0.00 - 2.75$	$2.75 - 5.51$	$5.51 - 8.26$	$8.26 - 11.02$	$11.02 - 13.78$	$13.78 - 16.53$	$16.53 - 19.29$	$19.29 - 22.05$

To evaluate the proposed quality measurement method, we perform the simulation comparing with RMSE, E, by using cross correlation.

$$RMSE = E = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1}\left(y[n] - x[n]\right)^2} \tag{6}$$

$$Cor\left(\epsilon, E\right) = \frac{\sum_{i=0}^{M-1}\left(\epsilon_i - \mu_\epsilon\right)\left(E_i - \mu_E\right)}{\sqrt{\sum_{i=0}^{M-1}\left(\epsilon_i - \mu_\epsilon\right)^2 \sum_{i=0}^{M-1}\left(E_i - \mu_E\right)^2}} \tag{7}$$

Where, $Cor(\epsilon, E)$ denotes the cross correlation of i-th frame. And the simulation is carried out on Gaussian noise channel, which has the mean of zero, and the standard deviation from 100 to 1,000.

As the high frequency band is more sensitive than low frequency band to noise. We experimentally select the 5th-band as optimal sub-band to modify the variance. The maximum and minimum value for converting onto normalized range is respectively determined as 32,767 and -32,768, since the range of audio signal quantized by 16-bits is from 32,767 to -32,768. As mentioned in section II, if the characteristics of signals could be known, we can calculate the reference value, r. In general, because the high frequency band has Laplacian distribution, the variance of normalized sub-band signals is nearby zero (That is, the distribution is concentrated on the mean value). For that reason, we determine r as 0.0001. Table 2 shows the audio quality after altering the variance of sub-band signals. Since the proposed method uses only specific frequency resource, the modified audio data has little quality distortion.

The simulation results after Gaussian noise addition is shown in Fig. 7, in terms of several standard deviations. The quality measurement is performed frame by frame. To compare with RMSE, we indicate the real range, not normalized range (Actually, the quality is represented from zero to one in the proposed audio quality measurement system). The circle dot denotes the result of RMSE, the squared dot is EVE, and the triangle dot is in case of no Gaussian noise. EVE shows the modality, which is proportional to RMSE. It means, the same as RMSE, EVE could estimate the quality distortion according to the strength of error. As shown in Fig. 8, the proposed system is high correlated with RMSE and table 3 represents the cross correlation between RMSE and EVE. Moreover, the proposed method is no reference technique, i.e., it does not require the original data for audio quality measurement.

TABLE II
THE MODIFIED AUDIO QUALITY (DB)

Music	Pop	Rock	Classic
SNR	30.34	46.17	33.72

TABLE III
CROSS CORRELATION BETWEEN RMSE AND EVE

Music	Pop	Rock	Classic
SNR	0.99	0.99	0.99

IV. CONCLUSION

In this paper, we proposed a new quality measurement technique for transmitted audio data by calculating the variance of intentionally modified sub-band signal. Through the simulations, we proved that the proposed enables to estimate the quality distortion, not requiring the original data at the receiver. As results, the proposed is a good alternative to the traditional quality measurement tool, RMSE.

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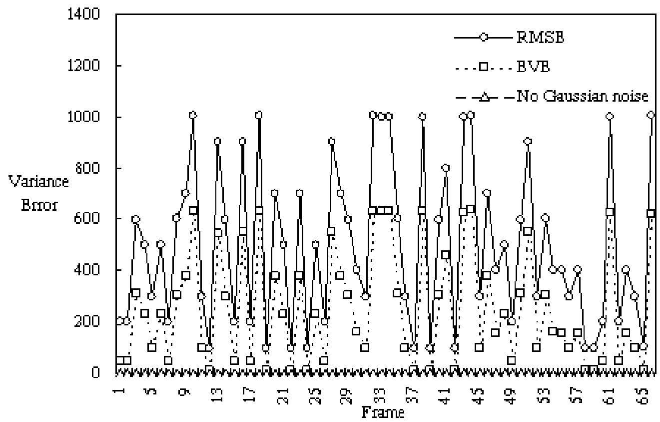

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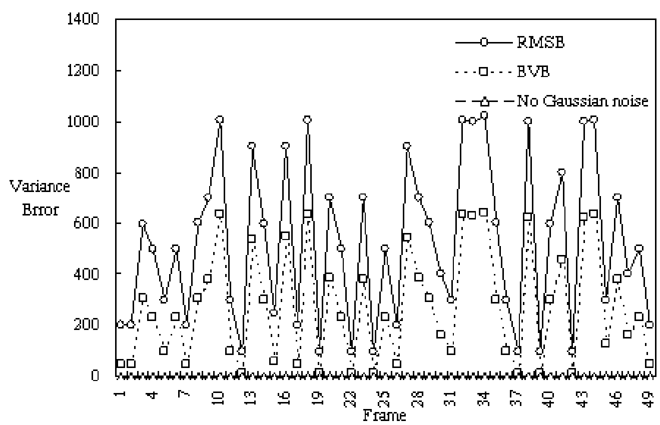

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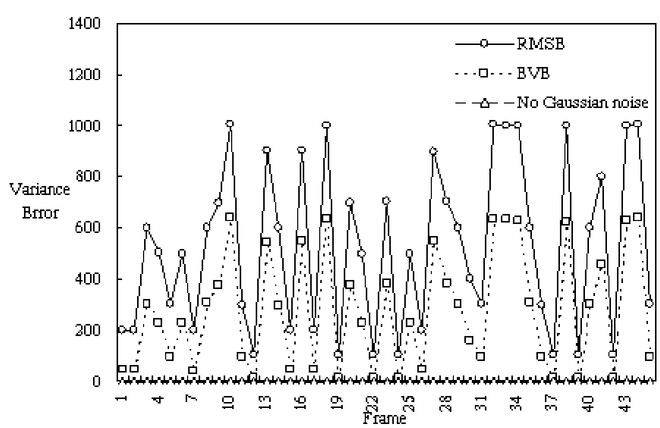
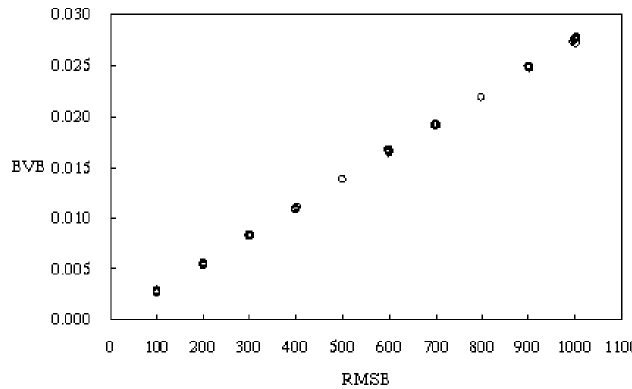


(c)

Fig. 7. Simulation result according to frame after Gaussian noise with several standard deviations (100 1,000)



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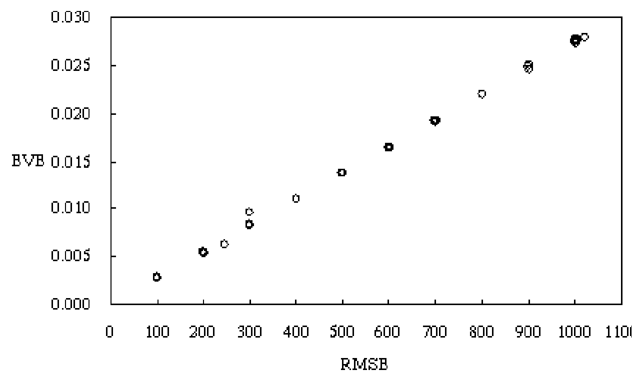

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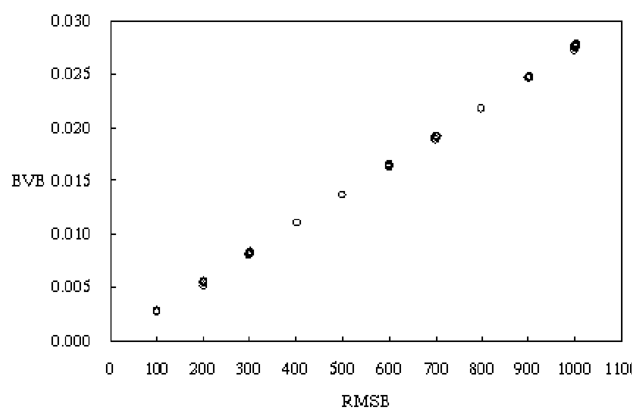

(c)

Fig. 8. The relation between RMSE and EVE